Listing 5.10 Computing a HMAC tag for a new token

```
package com.manning.apisecurityinaction.token;

import spark.Request;

import javax.crypto.Mac;
import java.nio.charset.StandardCharsets;
import java.security.*;
import java.util.*;

public class HmacTokenStore implements TokenStore {

    private final TokenStore delegate;    #A
    private final Key macKey;    #A

    public HmacTokenStore(TokenStore delegate, Key macKey)
{    #A
        this.delegate = delegate;
        this.macKey = macKey;
    }

    @Override
    public String create(Request request, Token token) {
        var tokenId = delegate.create(request, token);    #B
        var tag = hmac(tokenId);    #B

        return tokenId + '.' + Base64url.encode(tag);    #C
    }

    private byte[] hmac(String tokenId) {
        try {
            var mac =
Mac.getInstance(macKey.getAlgorithm());    #D
            mac.init(macKey);    #D
            return mac.doFinal(    #D

tokenId.getBytes(StandardCharsets.UTF_8));    #D
```